

Research Interests

Autonomous surgical robotics; vision-language-action models; embodied artificial intelligence; medical robotics; computer vision; robot learning; human-computer interaction.

Education

BEng	Durham University, University College <i>Electronics Engineering</i> • First Class Honours in Levels 1 and 2; cumulative GPA: 3.72/4.0 . Selected study in control, signal processing, digital electronics, communications, and electrical engineering.	2023–2027 United Kingdom
Study Abroad	University of Notre Dame <i>Electrical Engineering, International Study Year Abroad</i> • Coursework in circuit design, systems theory, semiconductor and optoelectronic devices, power electronics, and computer architecture.	2025–2026 United States

Research Experience

Summer Undergraduate Research Intern, The Chinese University of Hong Kong <i>Medical Robotics Group; Supervisor: Prof. Hongliang Ren</i> BiliVLA: Scene-Aware VLA for Autonomous ERCP Navigation	Jun–Aug 2026 Hong Kong SAR Research contributor
- Contributed to a system achieving 96.25% mIoU, 91.96% action precision, and 84.85% navigation success in ERCP phantom experiments, supporting reliable target understanding and closed-loop control. - Supported a two-stage pipeline combining grounding-enhanced supervised fine-tuning, GRPO reinforcement optimisation, scene-aware grounding, and safety-aware recovery; integrated the hardware platform, multimodal trajectories, and 3-DoF action definitions.	
Teleoperated Flexible Endoscopy Benchmark	<i>Dataset and systems contributor</i>
- Established a reproducible multimodal benchmark pipeline and collected an initial 150 synchronised endoscopic video and robot-state segments, resolving timestamp alignment across video, high-frequency robot states, and PDT hardware parameters. - Built cleaning, frame-extraction, YOLO ROI-cropping, and 224×224 image-action trajectory workflows; deployed the open-source $\pi_{0.5}$ policy as a baseline for VLA navigation experiments.	
Trileg-ESD: Vision-Guided Magnetic Soft Robot	<i>Research contributor</i>
- Advanced a miniature three-legged magnetic soft robot for auxiliary tissue manipulation, structuring controllable motion around squatting, leg lifting, rotation, crawling, and stance-recovery primitives. - Investigated a vision-language controller mapping current and recent endoscopic frames plus motion instructions to voltage commands for three electromagnetic coils.	
Google Summer of Code Mentor, International Neuroinformatics Coordinating Facility	2026 <i>Remote</i>
<i>Open-source neuroinformatics and computational neuroscience</i>	
Selected as an official GSoC mentor; evaluated proposals, reviewed code, and advised international contributors on reproducible scientific software and research-oriented architecture.	
MITACS Globalink Research Intern, McGill University, CIRMMT <i>Supervisor: Prof. Suresh Krishna</i>	Jun–Aug 2025 Montreal, Canada
Built and evaluated a Python-based, low-latency markerless motion-to-sound workflow using AI-assisted motion capture and Google MediaPipe for gesture-controlled music generation. - Contributed to experimental design, coding, analysis, and reporting across engineering, neuroscience, psychology, and human-computer interaction.	

Research Experience (continued)

HPC Summer Research Student, EPCC, University of Edinburgh <i>High-performance computing and scientific workflows</i> Applied parallel-computing, large-scale data-processing, and scientific-computing methods to engineering problems after intensive training in HPC systems and architectures.	Jun-Jul 2024 <i>United Kingdom</i>
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Selected Engineering Projects

Automated Material Mover Co-designed and prototyped an autonomous forklift system for industrial material transport under strict hardware constraints, integrating mechanical design, electronics, sensing, and iterative reliability testing.	<i>Durham Engineering Design Challenge,</i> <i>2023–2025</i>
IMechE Repeatable Vehicle Challenge Engineered, fabricated, calibrated, and repeatedly tested an autonomous vehicle, improving mechanical repeatability and circuit reliability through iterative debugging.	<i>Durham University,</i> <i>2023–2025</i>
Vienna Heart Hackathon Developed interdisciplinary medical-engineering concepts addressing cardiovascular-technology challenges and translated clinical needs into implementable engineering requirements.	<i>2025</i>

Technical Skills

Programming	Python; MATLAB; C/C++; Arduino programming; data processing and visualisation.
AI / Robotics	Computer vision; YOLO; vision-language-action models; reinforcement learning; multimodal trajectory datasets; robotic control and sensor integration.
Hardware / Tools	PCB design; analogue/digital electronics; microcontrollers; embedded systems; Multisim; LTspice; Cadence; SolidWorks; Git/GitHub; MediaPipe.
Computing	High-performance computing; parallel architectures; scientific computing; large-scale data processing and technical analysis.

Leadership and Service

Head of Design, DurHack <i>Durham University’s flagship annual hackathon</i> Led visual identity and user-facing digital communication with software and logistics teams.	Sep 2024–2025 <i>United Kingdom</i>
Student Representative and EDI Committee Member <i>Durham University Department of Engineering</i> Represented 150+ engineering students and served as Engineering Society Outreach Executive.	2023–2024 <i>United Kingdom</i>
Co-President, College Ultimate Frisbee <i>Durham University and University of Notre Dame</i> Managed logistics, recruitment, and multicultural team coordination; competed as a registered USA College player for Notre Dame.	2024–Present <i>UK / US</i>
President, Castle Pride <i>University College, Durham University</i> Led inclusion and community-building programmes and organised student discussions and events.	2023–2024 <i>United Kingdom</i>

Honours and Additional Information

Awards / Service	Two-time finalist, Durham University Annual Art Competition (2023, 2024); Open Day Representative, University College (2023).
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